

SPCE 5065

HOMEWORK #1

DUE: Friday 26 June 2026

All homework is due in Canvas by 11:59:59 pm MT (Colorado Time) on the due date indicated on the Syllabus schedule. No late homework will be accepted.

Students are allowed PARTIAL COLLABORATION on homework assignments. You are allowed to discuss qualitatively with other students the concepts in this course. However, copying or in any way using the written work of another person as well as relaying or receiving solutions via any means is strictly prohibited. The intent of this policy is to allow you to share ideas, discuss concepts, and clarify processes when needed. However, you must independently prepare the detailed solutions to homework problems and the Final Project.

Use course readings as a resource, but research additional sources as necessary to answer the questions. Please annotate your research with [1], [2], etc. and a sources page that describe where the information is located in AIAA format. Please email me with any questions or concerns.

1. Find an example of a spacecraft anomaly due to the space environment. Describe what happened and what improvements were made to avoid a recurrence of the problem.
2. Newton's Universal Law of Gravitation states:

$$F = G \frac{M_E m_s}{(R_E + h)^2}$$

Where:

G = Universal Gravitational Constant $\frac{Nkm^2}{kg^2}$

M_E = mass of the Earth (kg)

m_s = mass of the spacecraft (kg)

R_E = radius of the earth = 6378 km)

h = spacecraft orbital altitude (km)

- a) Write the centripetal acceleration of the satellite in terms of orbital velocity (km/s), R_E and h.
- b) **Derive** an expression for the orbital velocity as a function of altitude and μ , were

$$\mu = GM_E = 398600.5 \frac{km^3}{s^2}$$

- c) Graph velocity versus altitude
- d) **Derive** an expression for the period of the orbit as a function altitude and μ .

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e) Graph period versus altitude

3. For a satellite launched to an altitude of 800 km, is there any significant difference to the lifetime depending on the phase of the solar cycle at launch? Explain your answer.

4. An Earth Observation spacecraft with an optical payload is in a 350 km circular orbit. What are some of the major problems operators might expect from the space environment in this regime? What will help mitigate those risks? Be sure to include an overview of all effects covered on lesson 1, not just those in Chapter 1 of the textbook.

5. Why does the Earth's magnetic field drift? How do we know the magnetic field reverses periodically? When is the next one predicted to occur?

6. The solar constant at any distance from the sun is given by:

$$S(r) = S_e \left(\frac{au}{r} \right)^2$$

a) Determine an expression for the Solar Constant, $S(r)$, as a function of orbit eccentricity given:

$S(r)$ = Sun's irradiance at any distance from the sun

$S_e = 1366.1 \frac{W}{m^2}$ (sun's irradiance at 1 AU)

$au = 1 \text{ AU} = 149,597,871 \text{ km}$ (distance from the Earth to the Sun)

r = distance from the sun (AU)

b) Determine the maximum and minimum values of the Earth's solar constant

c) Graph the solar irradiance at the earth as a function of the day of year.

d) Calculate and plot the average solar irradiance along with the max and min values for each of the planets in our solar system. Use a log scale for the y axis.

7. Observations of Titan indicate that it has a period of 14.1 Earth days.

a) Determine the mass of Saturn if the semi-major axis of Titan's orbit is 1,110,781,765 m.

b) Find a published estimate of Saturn's mass and determine the percent difference between your calculation and the published mass.

c) Explain why there is a difference.

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8. A spacecraft is in an eccentric orbit about an asteroid with a semimajor axis of 1000 km. At a distance from the asteroid of 1500 km, the velocity has a magnitude of 10 m/s. Determine the mass of the asteroid.